

# Lesson 1: Reliability of Data

## Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

## Lesson focus

Students learn how scientists judge whether data can be trusted. They are introduced to uncertainty, precision, accuracy, resolution of measurement, random error, and systematic error before they begin planning and presenting investigations in the rest of the unit.

## Why this lesson comes first

This lesson prepares students for the practical and data-handling skills used throughout Lessons 2-8. Before students choose graph types, calculate means, draw best-fit lines, or evaluate outliers, they need to understand why measurements are never perfect and why repeated readings help us judge reliability.

## Learning objectives

By the end of the lesson, students should be able to:

- 1 Explain why measurements always have some uncertainty.
- 2 Calculate uncertainty from repeated readings using half the range.
- 3 Distinguish between precision and accuracy.
- 4 Explain what is meant by the resolution of a measuring instrument.
- 5 Describe random errors and systematic errors.
- 6 Suggest ways to improve the reliability of data.

## Key vocabulary

| Keyword          | Student-friendly definition                           |
|------------------|-------------------------------------------------------|
| Reliability      | How trustworthy or repeatable a set of results is     |
| Uncertainty      | The doubt about a measurement or result               |
| Range            | The difference between the largest and smallest value |
| Precision        | How close repeated measurements are to each other     |
| Accuracy         | How close a measurement is to the true value          |
| Resolution       | The smallest change a measuring instrument can detect |
| Random error     | Unpredictable variation between repeated measurements |
| Systematic error | An error that affects all results in the same way     |
| Mean             | The average value                                     |

## AQA-style key ideas

| Idea                                | What students need to understand                                                           |
|-------------------------------------|--------------------------------------------------------------------------------------------|
| Uncertainty                         | Whenever a measurement is made, there is always some uncertainty about the result obtained |
| Estimating uncertainty from repeats | Use the range of repeated measurements about the mean as a measure of uncertainty          |
| Simple uncertainty calculation      | Uncertainty = (maximum value - minimum value) / 2                                          |
| Precision                           | Precise measurements are close together                                                    |
| Accuracy                            | Accurate measurements are close to the true value                                          |
| Random error                        | Random errors cause readings to vary unpredictably                                         |
| Systematic error                    | Systematic errors shift readings in the same direction each time                           |
| Resolution                          | A measuring instrument with smaller divisions has better resolution                        |

## Equipment and resources

- Mini whiteboards or exercise books
- Projected examples of repeated readings
- Meter rulers
- Stopwatches
- A toy car, trolley, or ball
- Optional: different measuring instruments, for example a ruler marked in cm and a ruler marked in mm
- Optional: printed worksheet with data sets

## Starter: Which data set is most reliable?

Show students three sets of repeated measurements for the same experiment.

Example question:

*A student measures the time taken for a toy car to travel down a ramp.*

| Data set | Trial 1 / s | Trial 2 / s | Trial 3 / s | Mean / s |
|----------|-------------|-------------|-------------|----------|
| A        | 1.20        | 1.22        | 1.21        | 1.21     |
| B        | 1.05        | 1.47        | 1.12        | 1.21     |
| C        | 0.91        | 0.92        | 0.91        | 0.91     |

Ask students:

- 1 Which set of readings is most precise?
- 2 Which set has the biggest spread?
- 3 Which set might be inaccurate even though it is precise?
- 4 Why is it useful to repeat measurements?

Expected discussion:

- Data set A is precise because the readings are close together.
- Data set B is less precise because the readings are spread out.
- Data set C is precise, but could be inaccurate if the true fall time is closer to 1.21 s.
- Repeats help us spot variation and calculate a more reliable mean.

## Main teaching: Precision and accuracy

Introduce the difference between precision and accuracy.

| Situation                           | Meaning                                                              |
|-------------------------------------|----------------------------------------------------------------------|
| Accurate and precise                | Measurements are close together and close to the true value          |
| Precise but not accurate            | Measurements are close together but far from the true value          |
| Accurate on average but not precise | Measurements are spread out, but the mean is close to the true value |
| Neither accurate nor precise        | Measurements are spread out and far from the true value              |

Useful wording:

*Precision is about how close repeated measurements are to each other.  
Accuracy is about how close a measurement is to the true value.*

## Main teaching: Random and systematic errors

Introduce the two main types of error.

| Error type       | Meaning                                                         | Example                                                                |
|------------------|-----------------------------------------------------------------|------------------------------------------------------------------------|
| Random error     | Unpredictable variation between repeats                         | Different reaction times when starting or stopping a stopwatch         |
| Systematic error | A repeated error that shifts all readings in the same direction | A ruler starts from the wrong position or a balance is not set to zero |

Students should understand:

- Repeating measurements and calculating a mean can reduce the effect of random errors.
- Repeating measurements does not remove systematic errors.
- Systematic errors are improved by checking equipment and method.

## Main teaching: Resolution of measurement

Resolution means the smallest change a measuring instrument can detect.

Examples:

| Instrument                  | Resolution |
|-----------------------------|------------|
| Ruler marked every 1 cm     | 1 cm       |
| Ruler marked every 1 mm     | 1 mm       |
| Stopwatch reading to 0.01 s | 0.01 s     |
| Balance reading to 0.1 g    | 0.1 g      |

Key point:

*A smaller resolution allows more precise measurements, but it does not automatically make the results accurate.*

## Activity 1: Calculate uncertainty from repeated readings

Teach students to estimate uncertainty using:

$$\text{Uncertainty} = (\text{maximum value} - \text{minimum value}) / 2$$

Example:

| Trial | Time / s |
|-------|----------|
| 1     | 1.20     |
| 2     | 1.25     |
| 3     | 1.22     |

Step-by-step:

- 1 Maximum value = 1.25 s
- 2 Minimum value = 1.20 s
- 3 Range = 1.25 - 1.20 = 0.05 s
- 4 Uncertainty = 0.05 / 2 = 0.025 s

The result can be written as:

$$\text{Mean time} = 1.22 \text{ s} \pm 0.025 \text{ s}$$

For Year 9, students can also round this sensibly:

$$\text{Mean time} = 1.22 \text{ s} \pm 0.03 \text{ s}$$

## Activity 2: Practice questions

Students calculate the mean, range, and uncertainty for each data set.

| Data set | Trial 1 | Trial 2 | Trial 3 | Mean | Range | Uncertainty |
|----------|---------|---------|---------|------|-------|-------------|
| A        | 12.0    | 12.2    | 12.1    |      |       |             |
| B        | 8.4     | 9.1     | 8.7     |      |       |             |
| C        | 25.0    | 25.0    | 25.1    |      |       |             |
| D        | 3.20    | 3.65    | 3.25    |      |       |             |

Challenge questions:

- 1 Which data set is most precise?
- 2 Which data set has the largest uncertainty?
- 3 Which data set might need to be repeated?
- 4 What could cause the readings to be spread out?

### Activity 3: Mini practical

Students complete a short repeated-measurement task.

Possible options:

- 1 Measure reaction time using a dropped ruler.
- 2 Measure the distance travelled by a toy car after rolling down a small ramp.
- 3 Measure the bounce height of a ball.
- 4 Measure the length of the same object using different rulers.

Students collect at least 3 repeated readings.

| Measurement | Trial 1 | Trial 2 | Trial 3 | Mean | Range | Uncertainty |
|-------------|---------|---------|---------|------|-------|-------------|
|             |         |         |         |      |       |             |

Students then answer:

- 1 Were your results precise?
- 2 What was your uncertainty?
- 3 What random errors may have affected your results?
- 4 What systematic errors may have affected your results?
- 5 How could you improve the reliability of your data?

### Check for understanding

Ask students:

- 1 What does precision mean?
- 2 What does accuracy mean?
- 3 What is the resolution of a ruler marked every 1 mm?
- 4 Why can a result be precise but not accurate?
- 5 What is a random error?
- 6 What is a systematic error?
- 7 Why does taking repeats help with random error?
- 8 Why does taking repeats not fix systematic error?

### Plenary: Reliability exit ticket

Students complete:

- 1 A precise set of results is \_\_\_\_\_.
- 2 An accurate result is \_\_\_\_\_.
- 3 The uncertainty is calculated using \_\_\_\_\_.
- 4 One possible random error was \_\_\_\_\_.
- 5 One possible systematic error was \_\_\_\_\_.
- 6 To make my data more reliable, I could \_\_\_\_\_.

## Possible prep

Students are given three small data sets and must:

- 1 Calculate the mean.
- 2 Calculate the range.
- 3 Calculate the uncertainty using half the range.
- 4 Identify which data set is most precise.
- 5 Suggest one possible random error and one possible systematic error.

## Teacher notes

This lesson should stay practical and concrete. The aim is not to teach advanced uncertainty analysis, but to give students the language and habits they will need for the rest of the unit.

The most important links to later lessons are:

- 1 Repeats help students judge reliability.
- 2 Means are useful because repeated measurements vary.
- 3 Unusual results may be outliers, but should be checked before being ignored.
- 4 Graphs and conclusions are only as good as the quality of the data behind them.
- 5 Evaluations should mention specific random errors, systematic errors, and realistic improvements.

Common misconception:

Students often think precise means correct. Emphasize that precise measurements are close together, but they may still be inaccurate if there is a systematic error.